

Learning Management System

Maham Javed

April 2026

Registration No: 2025-GA-49

1. Introduction

The Learning Management System is a desktop application developed for the Department of Computer Science, UET Lahore. This system streamlines the management of final year projects by replacing traditional spreadsheet-based processes.

The system allows:

- Management of students and advisors
- Project creation and assignment
- Group formation
- Evaluation and marks management
- Report generation

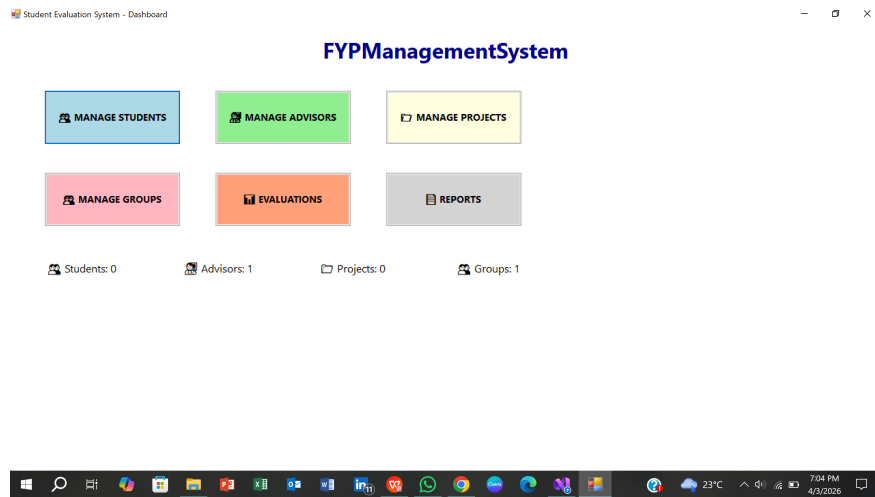


Figure 1: Dashboard Interface

2. Project Overview

This application is designed for the LMS committee to manage the entire lifecycle of final year projects.

Key Functions:

- Manage Students (Add, Update, Delete, View)
- Manage Advisors (Add, Update, Delete, View)
- Manage Projects (Add, Update, Delete, View)
- Create Groups
- Assign Students to Groups
- Assign Projects to Groups
- Assign Advisors to Projects
- Manage Evaluations
- Assign Marks to Groups
- Generate PDF Reports

3. Technologies Used

Component	Technology
Frontend	C# Windows Forms (.NET Framework 4.8)
Database	MySQL
Database Connectivity	MySQL Connector/NET
Reports	RDLC / DataGridView
Version Control	Git & GitLab

Table 1: Technologies Used

4. Database Design

Database Name: projectadb26

4.1 Tables Created

Table Name	Description
person	Stores personal information (name, email, contact)
student	Stores student details (registration number)
advisor	Stores advisor details (designation, salary)
project	Stores project details (title, description)
group	Stores group information
groupstudent	Maps students to groups
groupproject	Maps projects to groups
projectadvisor	Maps advisors to projects with roles
evaluation	Stores evaluation details
groupevaluation	Stores marks obtained by groups
lookup	Stores lookup values

Table 2: Database Tables

4.2 Lookup Table Data

Category	Values
GENDER	Male, Female
DESIGNATION	Professor, Associate Professor, Assistant Professor, Lecturer, Industry Professional
ADVISOR_ROLE	Main Advisor, Co-Advisor, Industry Advisor
STATUS	Active, InActive

Table 3: Lookup Table Values

5. Features Implemented

5.1 Student Management

- Add new students with personal information
- Update existing student records
- Delete students from the system
- View all students in a data grid

Manage Students

Student Information

First Name: Date of Birth: +

Last Name: Gender:

Email: Registration No:

Contact:

First Name	Last Name	Email	Contact	Date of Birth	Gender	Registration No
------------	-----------	-------	---------	---------------	--------	-----------------

Figure 2: Manage Students Form

5.2 Advisor Management

- Add advisors with designation and salary
- Update advisor information
- Delete advisors from the system

Manage Advisors

First Name: Designation:

Last Name: Salary:

Email:

Contact:

First Name	Last Name	Email	Contact	Designation	Salary
muham.javed	javed	muham@gmail.com	085955900	Lecturer	500000

Figure 3: Manage Advisors Form

5.3 Project Management

- Add new projects with title and description
- Update project details

- Delete projects from the system

The screenshot shows a web application window titled "Manage Projects". It contains two text input fields labeled "Title:" and "Description:". To the right of these fields are four buttons: "ADD" (green), "UPDATE" (yellow), "DELETE" (red), and "REFRESH" (grey). Below the input fields is a table with two columns, "Title" and "Description". The table is currently empty. The window is running on a Windows operating system, as indicated by the taskbar at the bottom.

Figure 4: Manage Projects Form

5.4 Group Management

- Create new groups
- Assign students to groups
- Assign projects to groups
- Delete groups

The screenshot shows a web application window titled "Manage Groups". It contains five buttons: "+ CREATE NEW GROUP" (green), "DELETE GROUP" (red), "ASSIGN STUDENTS" (blue), "ASSIGN PROJECT" (yellow), and "REFRESH" (grey). Below the buttons is a table with three columns: "Id", "Created_On", and "TotalStudents". The table contains four rows of data:

Id	Created_On	TotalStudents
24	4/3/2026	0
23	4/3/2026	2
22	4/3/2026	0
21	2/21/2026	0

The window is running on a Windows operating system, as indicated by the taskbar at the bottom.

Figure 5: Manage Groups Form

5.5 Evaluation Management

- Create evaluations (e.g., Proposal, Mid, Final)
- Assign marks to groups for each evaluation
- Update existing marks

Manage Evaluations

Evaluation Name: Total Marks: Weightage:

Name	TotalMarks	TotalWeightage
pr	15	10
q1	10	10

Figure 6: Manage Evaluation Form

5.6 Report Generation

- **Projects Report:** Shows projects with assigned advisors and students
- **Marks Report:** Shows marks obtained by each student in each evaluation
- **Students Report:** Shows all registered students

Generate Reports

of | 100% | Find | Next

An error occurred during local report processing.
The report definition for report 'ProjectReport' has not been specified.
Could not find file 'C:\Users\Wadullah\OneDrive\Desktop\FTPM\ManagementSystem\ProjectFiles\MSD\25_25GA49\bin\Debug\ProjectReport.rdlc'

Figure 7: Reports Form

6. Forms / Modules

Form Name	Purpose
DashboardForm	Main menu with navigation buttons
ManageStudentsForm	CRUD operations for students
ManageAdvisorsForm	CRUD operations for advisors
ManageProjectsForm	CRUD operations for projects
ManageGroupsForm	Group creation and management
AssignStudentsToGroupForm	Assign students to groups
AssignProjectToGroupForm	Assign projects to groups
ManageEvaluationsForm	Create and manage evaluations
AssignMarksForm	Assign marks to groups
ReportForm	Generate PDF reports
DatabaseHelper	Database connection and queries

Table 4: Forms and Modules

7. Steps to Run the Project

7.1 Prerequisites

- Visual Studio 2022 with .NET Framework 4.8
- MySQL Server & MySQL Workbench
- MySQL Connector/NET

7.2 Step-by-Step Instructions

Step 1: Restore Database

- Open MySQL Workbench
- Create database: `CREATE DATABASE projectadb26;`
- Run the provided SQL script

Step 2: Open Project

- Open MidDb26_25GA49.sln in Visual Studio

Step 3: Install Packages

- Open Package Manager Console
- Run: `Install-Package MySql.Data`

Step 4: Update Connection

- Open DatabaseHelper.cs
- Update password in connection string

Step 5: Build and Run

- Press F5 to run the application

8. Project Structure

```
MidDb26_25GA49/
DatabaseHelper.cs      # Database connection class
DashboardForm.cs       # Main dashboard
ManageStudentsForm.cs  # Student management
ManageAdvisorsForm.cs  # Advisor management
ManageProjectsForm.cs  # Project management
ManageGroupsForm.cs    # Group management
AssignStudentsToGroupForm.cs
AssignProjectToGroupForm.cs
ManageEvaluationsForm.cs # Evaluation management
AssignMarksForm.cs     # Marks assignment
ReportForm.cs          # Report generation
ProjectReport.rdlc     # RDLC report file
ProjectDataSet.xsd     # Dataset for report
Program.cs             # Entry point
```

9. GitLab Repository

The project is maintained on GitLab for version control and collaboration.

Repository URL:

https://gitlab.com/maham-ja/midddb26_25ga49

Collaborator Added:

samyan.uet@gmail.com (Teacher)

Git Commands Used:

```
git init
git add .
git commit -m "Final Project Complete"
git remote add origin https://gitlab.com/maham-ja/midddb26_25ga49.git
git push -u origin master
```


10. Conclusion

The Learning Management System successfully streamlines the management of final year projects at UET Lahore. The system provides:

- User-friendly interface
- Complete CRUD operations
- Efficient group and project assignment
- Evaluation and marks tracking
- Professional PDF reports

The system eliminates the need for spreadsheets and provides a centralized, efficient platform for the LMS committee.

11. Future Enhancements

- Add authentication system for different user roles
- Email notifications for students and advisors
- Export reports to Excel format
- Dashboard charts and analytics
- Student login portal

12. Declaration

I hereby declare that this project is my original work and has not been submitted for any other examination.